01 / DESIGN SWEEP BRIEF

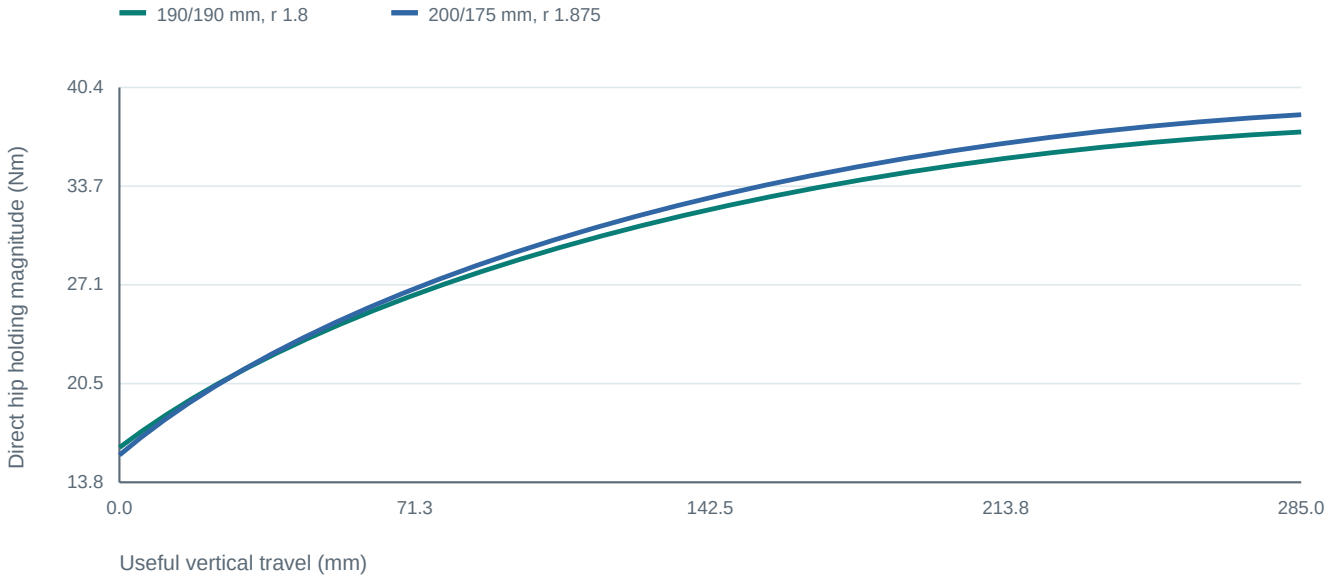
Another path-shaping variable

Slightly unequal links do not automatically improve vertical force or motor torque.

The current upper-long examples trade **0-4.9% less combined link length** for **1.1-5.0% more peak vertical-load hip torque** and slightly less useful angular ROM than the similar rounded equal-link examples. Combined length is only a size proxy: the longest individual link actually grows by 5-15 mm in these comparisons.

Sideways target mm	Equal links mm; guide	Upper-long mm; guide	Hip peak E /U Nm	Total length saved mm
25	185/185; r1.9	200/170; r2	38.39 /40.30	0
50	190/190; r1.8	200/175; r1.875	37.36 /38.52	5
75	200/200; r1.7	205/185; r1.75	37.09 /37.50	10
100	205/205; r1.6	215/175; r1.7	35.89 /36.89	20

Approximately 50 mm sideways cases: individual holding torque curves



Both families have 325 mm total/285 mm useful height and approximately matched sideways bands. Guide ratios and total sizes also differ, so this table compares practical candidates; it does not isolate link mismatch alone. Load is 110.3625 N vertical per leg, representing 45 kg equally shared over four corners, with massless moving links and no spring/inertia. These are not motor ratings.

Use equal 190/190 mm, r1.8 as the 50 mm baseline. Upper 200/175 mm,15:8 remains a useful geometry/guide-packaging option. It has not demonstrated a force advantage. Only optimizing equal and unequal families under identical constraints can compare their best achievable results.

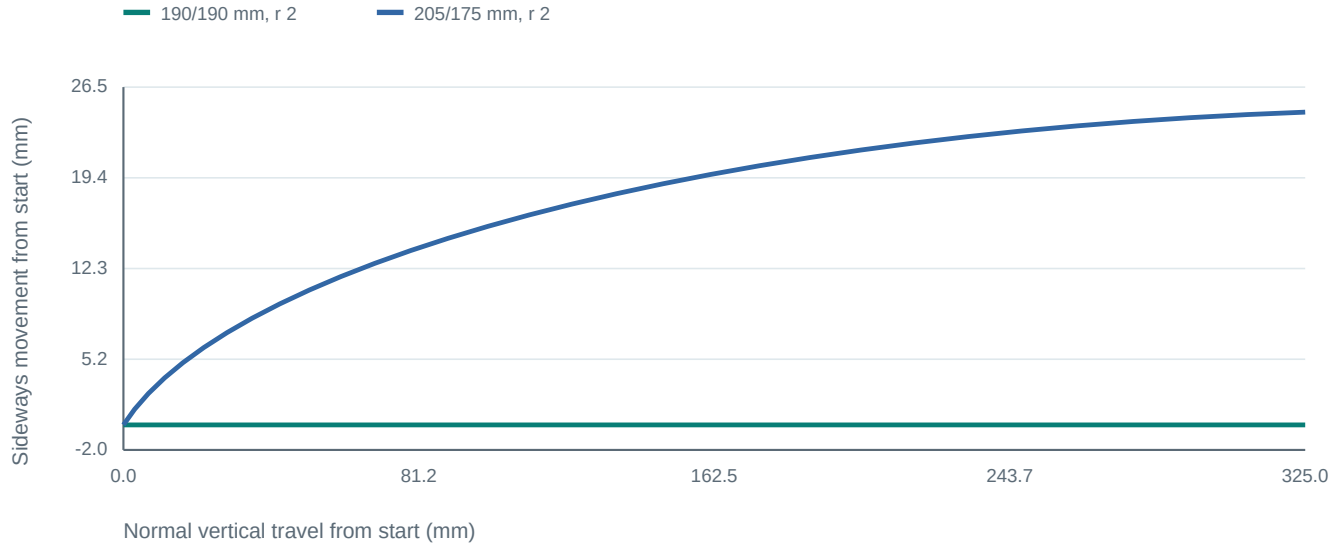
02 / DESIGN SWEEP BRIEF

Separate vertical and sideways motion

At guide 2:1, total length controls vertical kinematics and length difference controls sideways motion.

Let S=L2+L3, D=L2-L3 and t=q-pi. At guide r=2 exactly:
`x=D cos(t); z=S sin(t)`
`Jx=-D sin(t); Jz=S cos(t)`
At fixed S, changing D adds sideways movement without changing vertical height, vertical-load torque or useful ROM. Equal links give D=0 and a vertical line; unequal links give an ellipse. Horizontal loads add a different Jx torque term.

Sideways displacement versus vertical travel; independent axis ranges



Fixed S=380 mm, r2	Sideways mm	Hip peak Nm	Guide pitch surplus mm
190/190	0.000	41.236	10.0
205/175	24.538	41.236	25.0

A fixed 375 mm,15:8 comparison also illustrates the adjustment: equal 187.5/187.5 gives 31.34 mm sideways travel and 38.361 Nm peak; upper 200/175 gives 49.77 mm and 38.523 Nm. Both pass the assumed 10 mm guide pitch surplus. This is more sideways movement for about 0.4% more vertical-load peak at this particular ratio.

The equal 195/195,r1.7 fixed-size control reaches 324.800 mm by 175 degrees, just 0.200 mm below the numerical 325 mm threshold. It is retained as a failure rather than clipping stroke; that tiny threshold difference is not a robust practical advantage. The separate 185/185,r2 control has only 5 mm guide surplus, below the assumed 10 mm; it is an analytical control, not a fully feasible candidate.

03 / DESIGN SWEEP BRIEF

Your current design brief

Confirmed owner choices, historical requests and declared study assumptions remain distinct.

Topic	Current state
Mechanism	Jonathan variables/conventions; coupled 1DOF leg; compare equal, upper-long and lower-long links
Wheel /links	R200 mm; longer/shorter<=1.25; link-length ratio is separate from guide pulley ratio
Stroke /buffers	Approximately 325 mm total including buffers; 20+20 mm split is assumed, giving 285 mm useful
Path /inversion	alpha=phi0=0; curved sideways 25/50/75/100 mm; ideal height inverse through full fold
Cancelled method	No R2 or fixed 15-degree straight-line/corridor ranking; 10-15 degree line was an earlier preference
Loads /motors	Robot 40-45 kg; 3g/4g held-load sensitivity; peak/RMS torque, speed and acceleration requested
Springs /checks	Refit existing spring topologies on finalists; compare Jonathan V2, Andy 2D and actual Pymunk
Delivery /history	Offline HTML/PDF, labelled plots/GIFs, app MD and revision history; keep reports separate from Motion Lab tooling

The screenshot originally showed q100-165 degrees. Recent compact examples extend normal motion to almost 175, with a 5 degree fold guard. Those are study choices, not established hardware stops. Full ideal inversion is checked on 100-260; any newly lowered normal start must also expand the full domain consistently. For example, 190/190 mm,r1.8 supplies only 269.21 mm over 100-165 degrees. The compact 325 mm result relies on the larger angular window.

Unequal links shift the fold wheel-centre position by L2-L3, but do not solve R200 wheel/link overlap. Upper-long folds put C inside upper AB; equal folds put C at A. Both need actual axial/chassis/hub/tire packaging. Permitted sideways movement is not measured compliance. Prior spring/motor cases use other geometries; do not transfer their ratings. Dual-rope remains IMPOSSIBLE /rejected.

04 / DESIGN SWEEP BRIEF

A fair next optimization sweep

Proposed ranges and method only; this brief does not run a new large grid.

Variable /scenario	Proposed starting study
Total /imbalance	S350-600 mm; d=(L2-L3)/S in[-1/9,+1/9], including exact d0; record actual individual bounds
Guide ratio	r1.2-2.6; compare sensible rational/tooth realizations afterwards; rb60 mm is an assumption
Operating window	Normal start 100-140; caps 165/170/175 as separate scenarios; optional 95-degree start sensitivity
Height /lateral	325 mm total,20+20 mm reserves; 25/50/75/100 +/-2.5 mm lateral bands, both directions labelled
Fixed architecture	alpha=phase=0; R200; original source conventions; equal/upper-long/lower-long families
Later physical stage	Measured axial layout, pulley outer sizes, moving mass/inertia, spring route/preload and motor gearing

Hard filters: owner length bound, exact total stroke and reserves, positive full-domain Jz with declared5 mm/rad conditioning threshold, no normal sideways reversal, guide pitch allowance and real clearances when supplied. Do not invent missing axial dimensions. Classify geometry-only results separately.

Keep separate objectives: peak hip torque, RMS torque for the same motion/duty, maximum link and actual swept envelope, useful ROM, joint speed/acceleration and side-load robustness. Use a Pareto set: retain designs not beaten in every chosen objective. No arbitrary blended weights; H/peak torque duplicates peak ranking when every candidate has the same H.

Sweep broadly, retain equal controls in every scenario, then refine selected tradeoffs continuously and round guide hardware. Recheck tolerances and active bounds; verify original2D math and actual Pymunk geometry. Finally refit springs and evaluate a common smooth lift time with moving masses and gearing. Impact simulation needs actual event/stroke/contact inputs; 3g/4g snapshots do not specify impact acceleration or damping.

Torque is not energy efficiency: ideal lift work is 31.453 J per leg for the common 285 mm/110.3625 N case. Relative-knee torque=-tau_hip/r and speed=-r times hip speed; half torque applies only at r2. Actual motor losses, thermal duty and compliance require their physical inputs. Source 14f53f1; workspace scope; canonical physics NOT APPLICABLE. Reproduce with compare_link_lengths.py and verify_sweep_brief.cjs; detailed Markdown retains the copyable next-sweep prompt.

13 analytical cases and 4,838 independent originalV2/Andy 2D scalar comparisons pass; maximum error 1.705e-13 mm/Nm. No new Pymunk, loaded contact, spring or hardware test is claimed by this brief.